

MSAI 699 Capstone Project

Capstone Project Proposal:

An Intelligent Multi-Agent Retrieval-Augmented Generation

System for Evidence-Based Cognitive Health Education

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An Intelligent Multi-Agent Retrieval-Augmented Generation (RAG) System for Evidence-Based Cognitive Health Education.

Problem Statement

Cognitive health disorders — including Alzheimer's disease, mild cognitive impairment (MCI), anxiety, and depression — affect more than one billion people globally, yet access to accurate, personalised, and evidence-based educational resources remains severely limited. General-purpose search engines surface unvetted, often contradictory information, while clinical materials are locked behind paywalls or written for specialist audiences. The result is a critical gap: patients, caregivers, and practitioners cannot efficiently obtain reliable guidance tailored to specific cognitive health contexts.

This project addresses that gap by engineering an intelligent multi-agent system that combines Retrieval-Augmented Generation (RAG) with coordinated AI agents. The system will retrieve, synthesise, and deliver precise, citation-grounded cognitive health information through a conversational interface — effectively functioning as a trusted, always-available educational companion.

Motivation & Impact

The motivation stems from three converging forces: the accelerating prevalence of cognitive decline in aging populations, the democratising potential of large language models (LLMs), and persistent inequities in health literacy. Building a RAG-based educational tool is technically significant because RAG mitigates hallucination — the foremost risk of deploying LLMs in health-adjacent domains — by anchoring responses to retrieved, authoritative documents.

Anticipated Impact

Patients & Caregivers On-demand, plain-language explanations of diagnoses, medications, and lifestyle interventions drawn from peer-reviewed sources.

Clinicians

Rapid synthesis of recent literature, clinical guidelines, and trial data to support evidence-based decision-making.

AI Research	A reproducible multi-agent RAG architecture demonstrating safe, domain-specific LLM deployment with auditable retrieval chains.
Health Equity	Open access design reduces the information asymmetry between well-resourced and underserved communities.

Research Questions & Hypothesis

Primary Research Question

RQ1

Can a multi-agent RAG system produce cognitive health educational responses that are more accurate, more grounded in evidence, and perceived as more trustworthy than responses generated by a baseline single-agent LLM without retrieval augmentation?

Secondary Research Questions

RQ2

What retrieval strategies (dense, sparse, hybrid) maximise relevance and factual fidelity in the cognitive health domain?

RQ3

How does agent specialisation (retrieval, reasoning, synthesis, fact-checking) affect overall response quality and latency?

RQ4

What prompt engineering and guardrail patterns most effectively prevent the system from producing harmful or misleading health information?

Hypothesis

A multi-agent RAG architecture — in which a retrieval agent surfaces domain-specific literature, a reasoning agent synthesises findings, and a validation agent cross-checks claims against authoritative sources — will outperform a single-agent baseline on RAGAS faithfulness, answer relevancy, and expert evaluation scores by at least 15%, while maintaining acceptable response latency for interactive use (< 8 seconds per query).

Proposed AI Techniques & Datasets

Core AI Techniques

Retrieval-Augmented Generation (RAG): Hybrid retrieval combining dense vector search (FAISS / Chroma) with sparse BM25 ranking. A re-ranking layer (Cohere Rerank or cross-encoder) ensures top-k passages are maximally relevant before generation.

Multi-Agent Orchestration: LangGraph or AutoGen framework managing four specialised agents: Retrieval Agent, Reasoning Agent, Validation/Fact-Check Agent, and Response Synthesis Agent.

Large Language Models: GPT-4o or Claude 3.5 Sonnet as the backbone LLM; Sentence-BERT / text-embedding-3-large for document and query embeddings.

Prompt Engineering & Guardrails: Chain-of-thought prompting, constitutional AI principles, and rule-based output filters to enforce safe, scope-bounded responses.

Evaluation Framework: RAGAS metrics (faithfulness, answer relevancy, context precision/recall) plus human expert evaluation on a 100-question benchmark set.

Datasets & Knowledge Sources

PubMed / MEDLINE Peer-reviewed biomedical abstracts and full-text articles on cognitive disorders, neuroscience, and mental health interventions.

Alzheimer's Assoc.	Clinical practice guidelines, caregiver resources, and dementia-care protocols from the Alzheimer's Association (alz.org).
NIH MedlinePlus	Plain-language health articles covering cognitive health, aging, and neurological conditions.
Cochrane Reviews	Systematic reviews and meta-analyses on pharmacological and non-pharmacological cognitive interventions.
CDC / WHO Reports	Population-level epidemiological data on dementia, depression, and anxiety prevalence.
Custom Benchmark	100 expert-curated Q&A pairs across five cognitive health sub-domains for evaluation.

Expected Challenges & Ethical Considerations

Technical Challenges

Hallucination Mitigation: Despite RAG, LLMs may still generate plausible but unsupported claims.

Mitigation: mandatory citation anchoring and cross-agent fact verification.

Retrieval Quality: Medical literature varies widely in quality and recency. Mitigation: sourcewhitelisting, recency weighting, and metadata filtering.

Agent Coordination Overhead: Multi-agent pipelines introduce latency. Mitigation: asynchronousagent invocation and caching of common retrievals.

Domain Specificity vs. Generality: Balancing deep medical relevance with accessible, lay-audienceexplanations. Mitigation: tiered response modes (patient / clinician).

Ethical Considerations

Scope of Practice: The system is explicitly positioned as an educational tool, not a clinical decision support system. Every response will include a disclaimer discouraging substitution for professional medical advice.

Bias & Health Equity: Training corpora may under-represent minority populations. Bias audits will be conducted across demographic sub-groups in the benchmark evaluation.

Privacy: No personal health information (PHI) will be collected or stored. The system operates on de-identified, publicly available literature only.

Transparency: All retrieved sources will be cited inline, enabling users to verify claims independently and fostering health literacy rather than dependency.

Misuse Prevention: Guardrails will block queries seeking specific diagnoses, prescription guidance, or crisis-intervention advice, redirecting users to appropriate professional resources.

Deliverables & Evaluation Criteria

Deliverable 1 Functional multi-agent RAG application deployed on Replit (Python / FastAPI backend, React/Streamlit frontend) with a live demo URL.

Deliverable 2	Vector knowledge base ingested from the five datasets listed above, indexed with FAISS or ChromaDB.
Deliverable 3	Evaluation report: RAGAS metrics vs. baseline LLM across 100-question benchmark; latency profiling; expert review summary.

Deliverable 4	Project report documenting architecture, design decisions, limitations, and future work (APA 7th edition).
Evaluation Rubric Alignment	
Clarity Well-scoped problem statement, precise RQs, and clear architecture narrative.	
Feasibility	Uses open-source frameworks (LangChain/LangGraph, FAISS, Chroma) deployable on Replit; public datasets; 12-week delivery timeline.
Originality	Novel combination of multi-agent orchestration and RAG in the underexplored cognitive health education domain.
AI Relevance	Directly applies LLMs, embedding models, vector retrieval, and agent frameworks — core MSAI curriculum competencies.